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In [24]: # =====
# Task 5 - Exploratory Data Analysis
# =====

import os
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Plot style
sns.set_style("whitegrid")

# Create graphs folder
os.makedirs("graphs", exist_ok=True)

# Load dataset
df = pd.read_csv("Titanic-Dataset.csv")

# -----
# Basic Information
# -----

print("="*60)
print("FIRST 5 ROWS")
print("="*60)
print(df.head())

print("\n")
print("="*60)
print("DATASET INFORMATION")
print("="*60)
df.info()

print("\n")
print("="*60)
print("STATISTICAL SUMMARY")
print("="*60)
print(df.describe())

print("\n")
print("="*60)
print("MISSING VALUES")
print("="*60)
print(df.isnull().sum())

print("\n")
print("="*60)
print("SURVIVAL COUNT")
print("="*60)
print(df["Survived"].value_counts())

print("\n")
print("="*60)
print("PASSENGER CLASS COUNT")
print("="*60)
print(df["Pclass"].value_counts())

print("\n")
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print("="*60)
print("GENDER COUNT")
print("="*60)
print(df["Sex"].value_counts())

# =====
# 1. Age Distribution
# =====

plt.figure(figsize=(8,5))
plt.hist(df["Age"].dropna(), bins=30)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.savefig("graphs/01_Age_Distribution.png", dpi=300, bbox_inches="tight")
plt.show()

# =====
# 2. Age Boxplot
# =====

plt.figure(figsize=(8,5))
sns.boxplot(x=df["Age"])
plt.title("Age Boxplot")
plt.savefig("graphs/02_Age_Boxplot.png", dpi=300, bbox_inches="tight")
plt.show()

# =====
# 3. Survival by Gender
# =====

plt.figure(figsize=(6,5))
sns.countplot(x="Sex", hue="Survived", data=df)
plt.title("Survival by Gender")
plt.savefig("graphs/03_Survival_by_Gender.png", dpi=300, bbox_inches="tight")
plt.show()

# =====
# 4. Survival by Passenger Class
# =====

plt.figure(figsize=(6,5))
sns.countplot(x="Pclass", hue="Survived", data=df)
plt.title("Survival by Passenger Class")
plt.savefig("graphs/04_Survival_by_Class.png", dpi=300, bbox_inches="tight")
plt.show()

# =====
# 5. Fare Distribution
# =====

plt.figure(figsize=(8,5))
sns.histplot(df["Fare"], bins=40)
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Count")
plt.savefig("graphs/05_Fare_Distribution.png", dpi=300, bbox_inches="tight")
plt.show()

# =====
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# 6. Pairplot
# =====

pair = sns.pairplot(
    df[["Age", "Fare", "Pclass", "Survived"]].dropna(),
    hue="Survived"
)
pair.savefig("graphs/06_Pairplot.png", dpi=300)

# =====
# 7. Correlation Heatmap
# =====

plt.figure(figsize=(8,6))
numeric_df = df.select_dtypes(include=["number"])
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.savefig("graphs/07_Correlation_Heatmap.png", dpi=300, bbox_inches="tight")
plt.show()

# =====
# 8. Scatter Plot
# =====

plt.figure(figsize=(8,5))
sns.scatterplot(x="Age", y="Fare", hue="Survived", data=df)
plt.title("Age vs Fare")
plt.savefig("graphs/08_Age_vs_Fare.png", dpi=300, bbox_inches="tight")
plt.show()

print("\n")
print("=*60)
print("EDA COMPLETED SUCCESSFULLY")
print("=*60)
print("Graphs have been saved in the 'graphs' folder.")
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FIRST 5 ROWS

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	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

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DATASET INFORMATION

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<class 'pandas.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	str
4	Sex	891 non-null	str
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	str
9	Fare	891 non-null	float64
10	Cabin	204 non-null	str
11	Embarked	889 non-null	str

dtypes: float64(2), int64(5), str(5)

memory usage: 83.7 KB

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STATISTICAL SUMMARY

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	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	
std	257.353842	0.486592	0.836071	14.526497	1.102743	
min	1.000000	0.000000	1.000000	0.420000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	

max	891.000000	1.000000	3.000000	80.000000	8.000000
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	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

MISSING VALUES

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

SURVIVAL COUNT

Survived	
0	549
1	342

Name: count, dtype: int64

PASSENGER CLASS COUNT

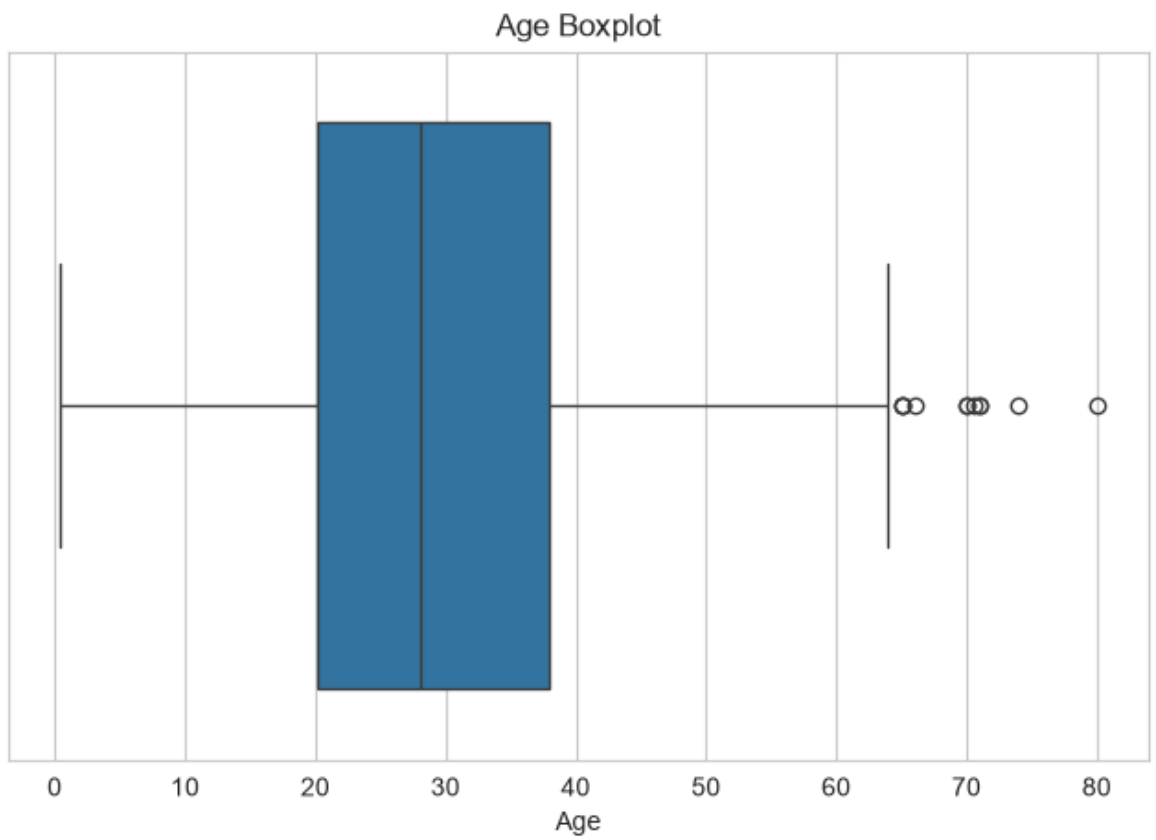
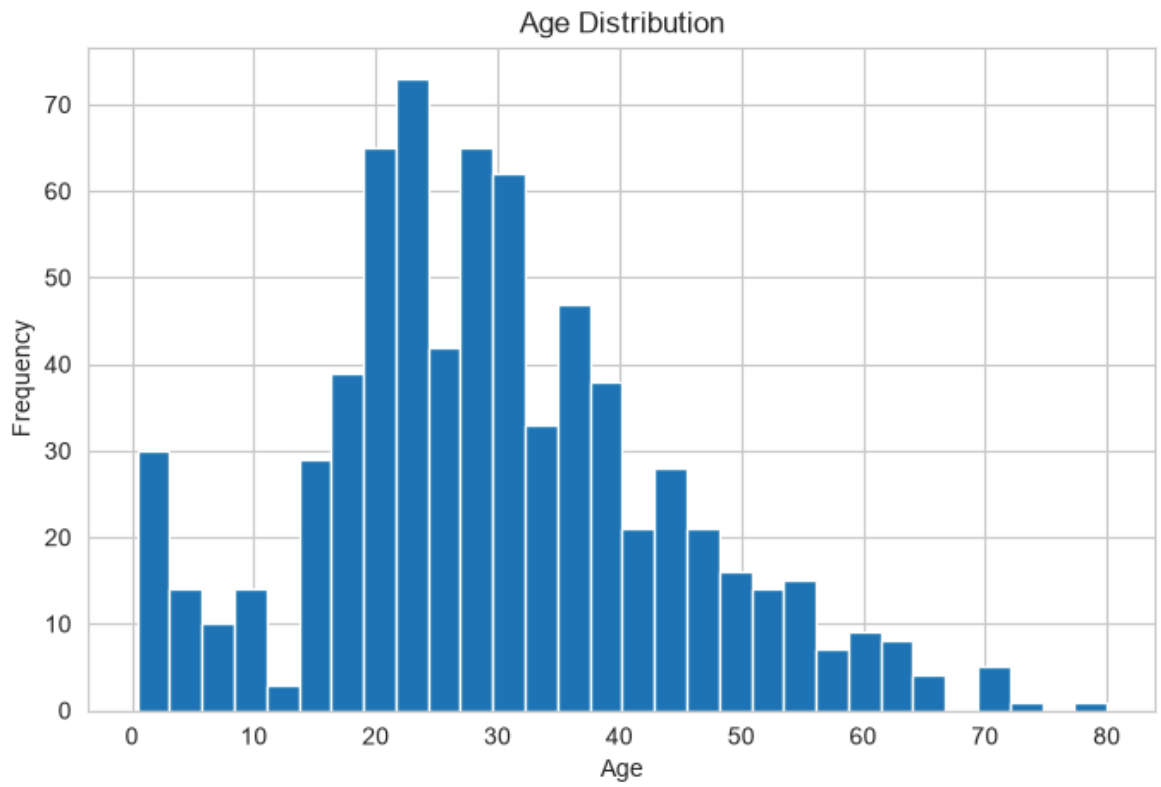
Pclass	
3	491
1	216
2	184

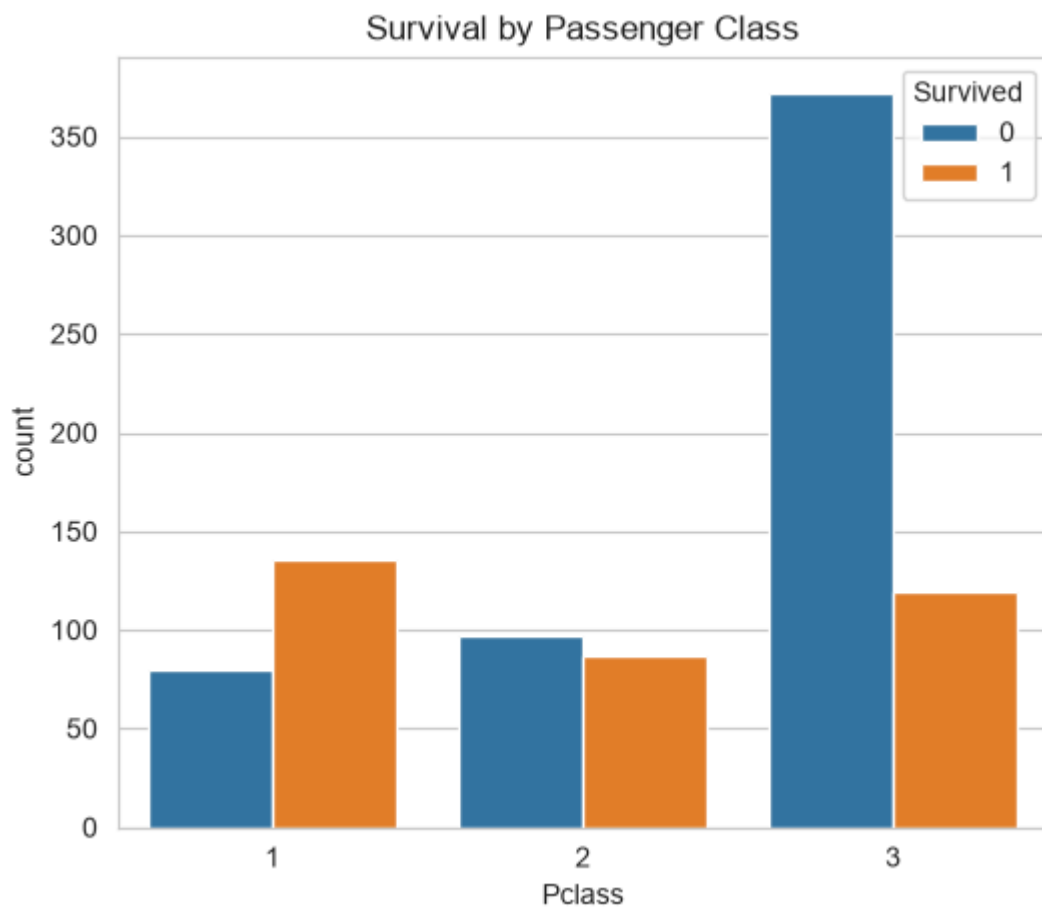
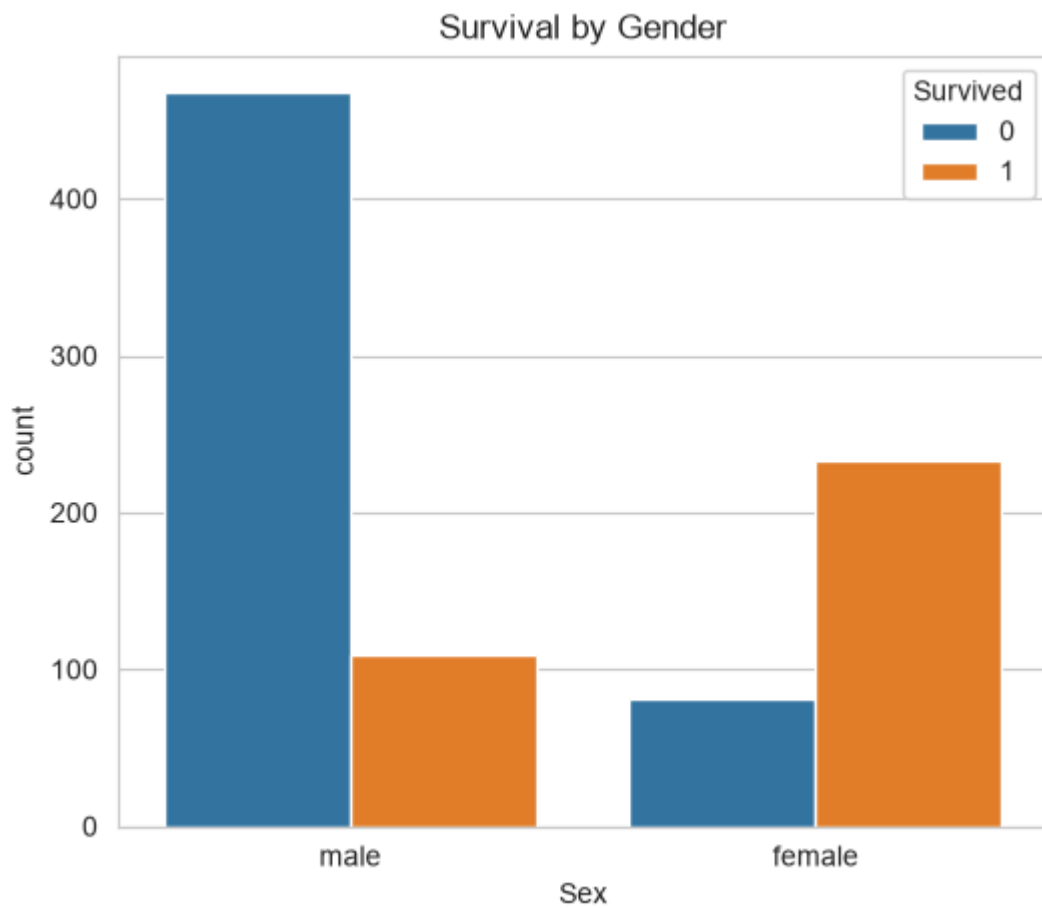
Name: count, dtype: int64

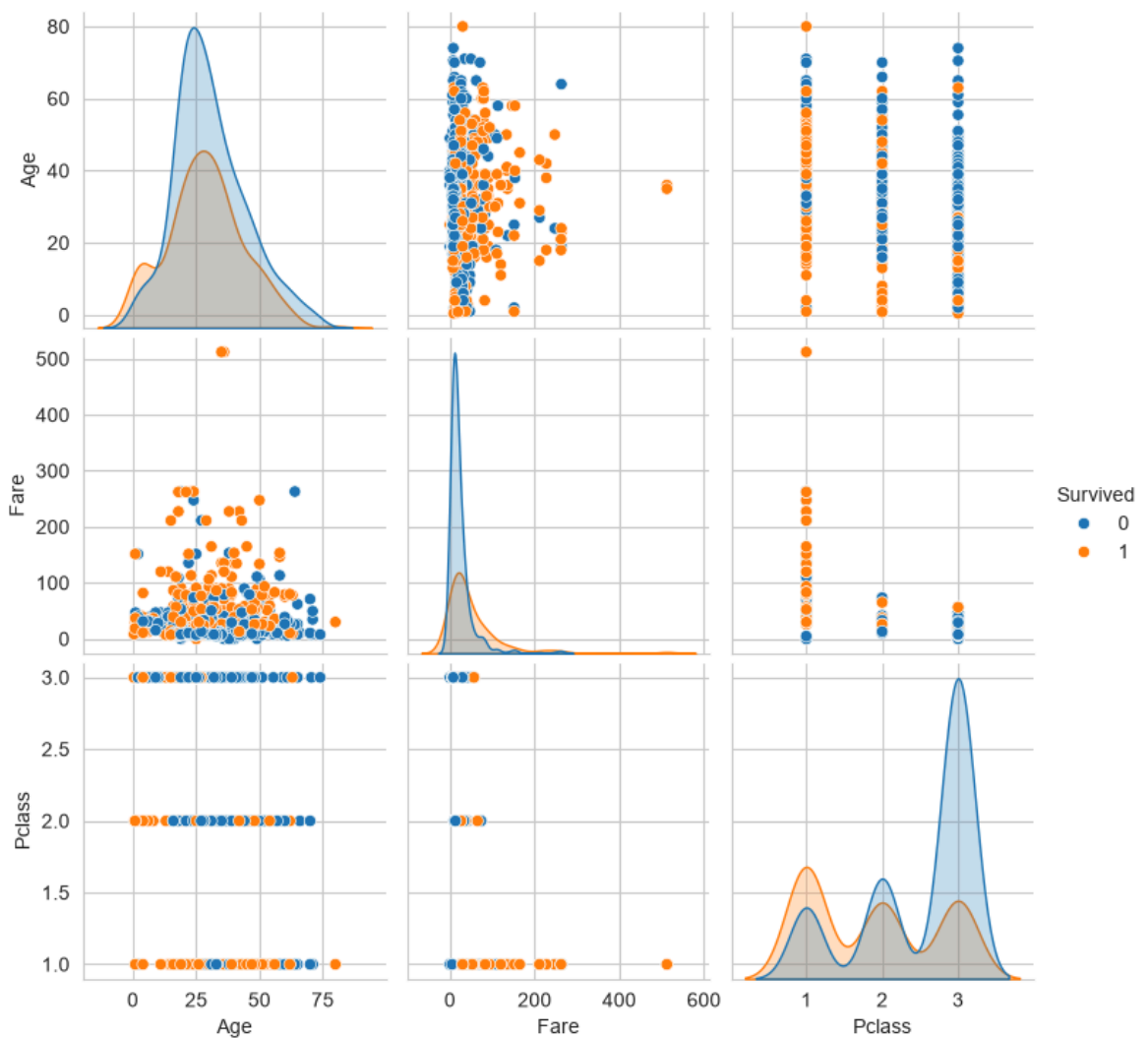
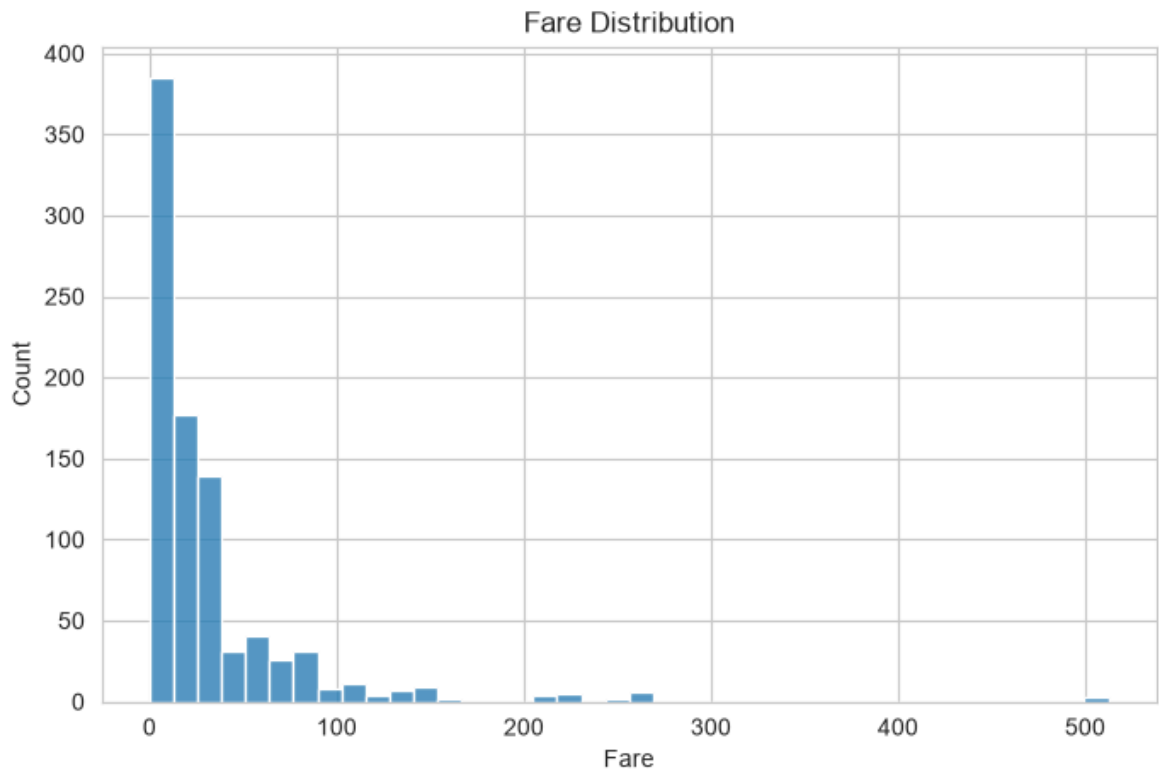
GENDER COUNT

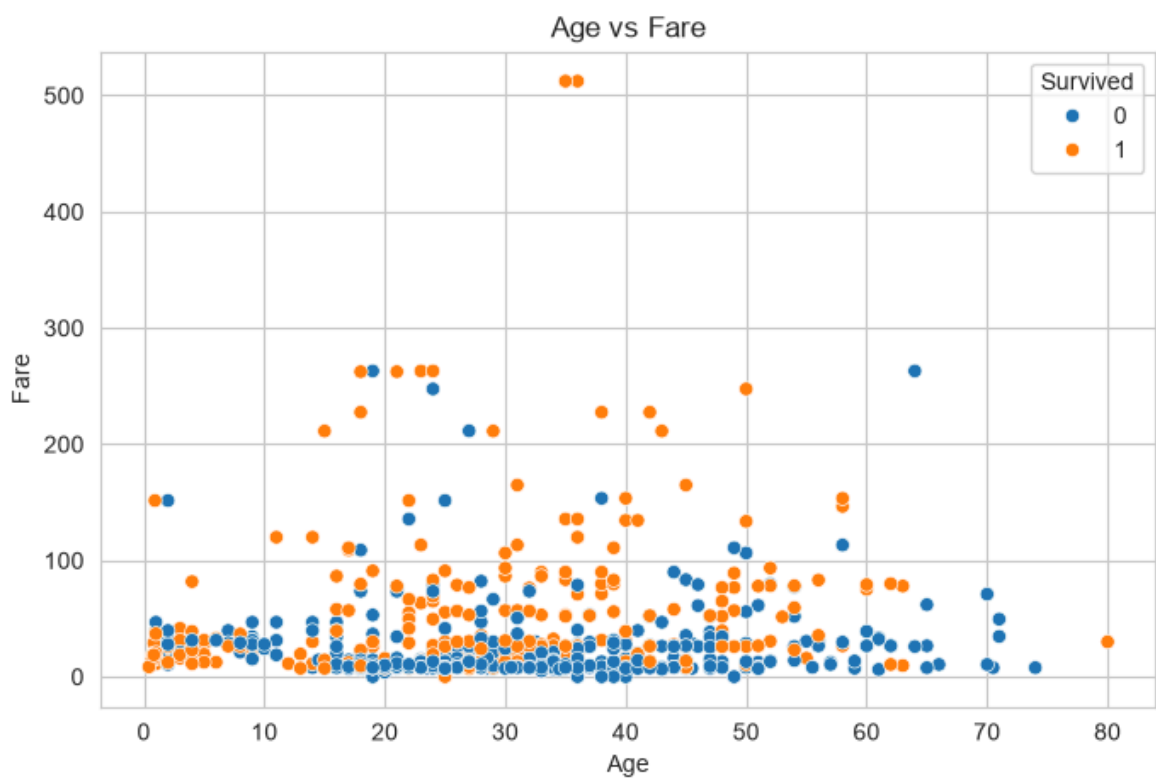
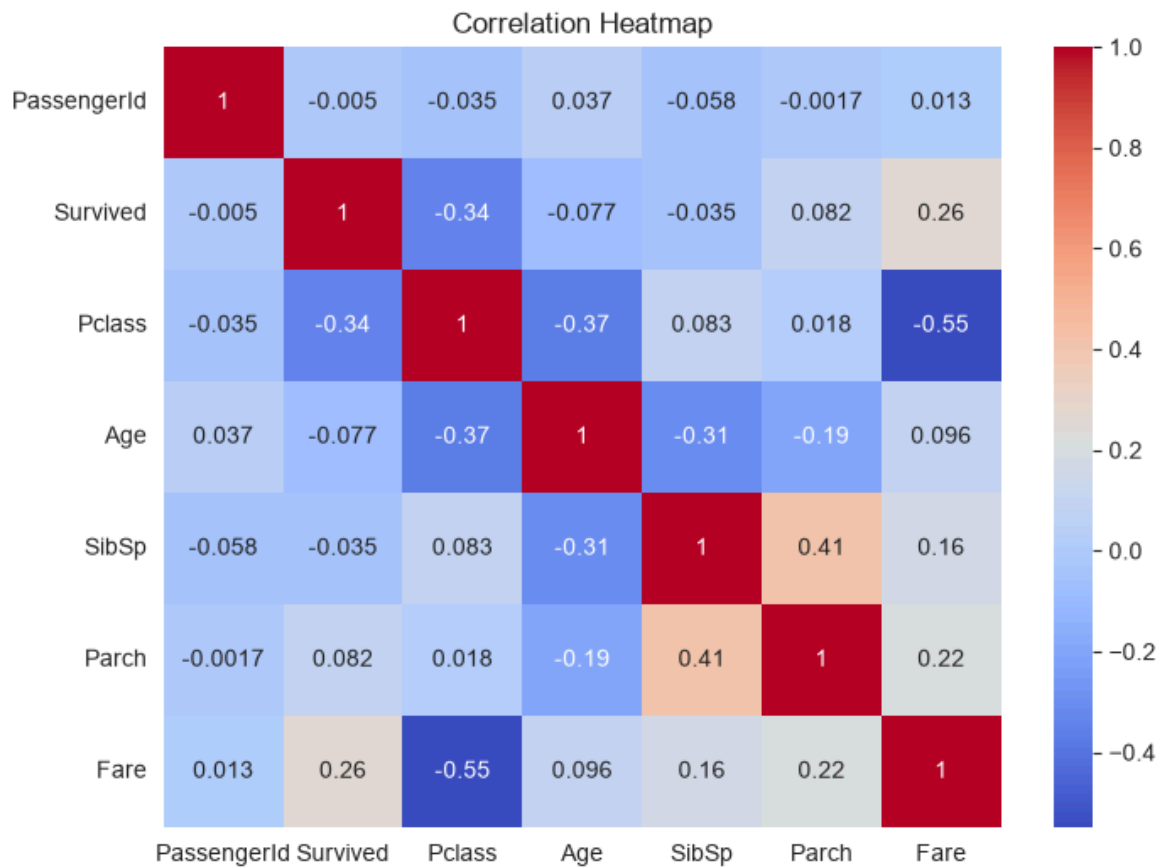
Sex	
male	577
female	314

Name: count, dtype: int64









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EDA COMPLETED SUCCESSFULLY
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Graphs have been saved in the 'graphs' folder.
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In []:

Final Observations

Missing Values

- The Age column contains missing values.
- The Cabin column has the highest number of missing values.
- The Embarked column has only a few missing values.

Age Distribution

- Most passengers were between 20 and 40 years old.

Survival Analysis

- Female passengers had a higher survival rate than male passengers.
- First-class passengers survived more frequently than second- and third-class passengers.

Fare Analysis

- Most passengers paid lower fares.
- A few passengers paid very high fares.

Correlation

- Passenger class and fare show a noticeable relationship with survival.

Conclusion

Exploratory Data Analysis revealed that gender, passenger class, and fare significantly influenced survival. Missing values in Age and Cabin should be handled before building a machine learning model.